

Econometrics II

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Problem Set 6

Due: March 25, 2026

Question 1.

Both sets of X 's are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY . Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). The columns of Y and X_{1t} indicate observations over time, ten periods in all (so in period 10 it is a static problem). The only uncertainty from the individual's perspective is on their error term. In other words, individuals have perfect foresight about their future health. The error structure is Type 1 extreme value. **Use "dataHW6 Problem1.mat" to complete this exercise. The purpose of this problem is to estimate a simple dynamic discrete choice problem. The data contain information on the choices (the Y 's), characteristics that do not vary over time (the X_1 's), and characteristics that do vary over time (the X_{1t} 's). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column X_1 allows for choice specific constants, while the second column of X_1 is an indicator for gender. X_{1t} is a time-varying measure of health.**

Both sets of X 's are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY . Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). The columns of Y and X_{1t} indicate observations over time, ten periods in all (so in period 10 it is a static problem). The only uncertainty from the individual's perspective is on their error term. In other words, individuals have perfect foresight about their future health. The error structure is Type 1 extreme value.

(a) Estimate the dynamic discrete choice model including the discount factor.

Solution.

Table 1: DDC Estimates

Parameter	
<i>Flex-Time</i>	
$\beta_{0,1}$	-0.961
$\beta_{Gender,1}$	0.368
$\beta_{Health,1}$	0.517
<i>Part-Time</i>	
$\beta_{0,2}$	-.051
$\beta_{Gender,2}$	0.687
$\beta_{Health,2}$	0.759
<i>Full-Time</i>	
$\beta_{0,3}$	-0.412
$\beta_{Gender,3}$	1.029
$\beta_{Health,3}$	0.992
δ	0.376
β	0.896

(b) What variation in the data allows the discount factor to be identified?

The discount factor is identified via the health state. It is the only time variant characteristic in X_{1t} , so it will pin down β .

(c) What variation in the data allows the switching cost to be identified?

Switching cost is identified via the constant, the only characteristic which is choice specific in X_1 . Gender is not something which we are assuming can change in this model, thus it must be the constant identifying the cost δ .

Question 2.

Individuals only have expectations over future values of X_{1t} . Given X_{1t} and Y_t , the transition is:

$$X_{1t+1} = \alpha_0 + \alpha_1 X_{1t} + \alpha_2(Y_t = 2) + \alpha_3(Y_t = 3) + \epsilon_{t+1}, \quad Y_t > 0$$

where $\epsilon_{t+1} \sim \mathcal{N}(0, \sigma)$ is unknown to the individual until $t + 1$. Use `dataHW6_Problem2.mat` to complete this exercise. The setup is identical to Problem 1. The data contain information on the choices (the Y 's), characteristics that do not vary over time (the X_1 's), and characteristics that do vary over time (the X_{1t} 's). The context is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flextime, part-time, or full-time participation. The first column of X_1 allows for choice-specific constants; the second column is an indicator for gender. X_{1t} is a time-varying measure of health. Both sets of X 's are individual-level characteristics, so coefficients on these variables vary by choice. There is a cost to switching labor supply that is common across all switches. The choice in period 0 is given by LY . Once an individual retires (chooses option 0), no further choices are made (if $Y_{it} = 0$ then $Y_{it+1} = 0$).

Individuals only have expectations over future values of X_{1t} . Given X_{1t} and Y_t , the transition is:

$$X_{1t+1} = \alpha_0 + \alpha_1 X_{1t} + \alpha_2(Y_t = 2) + \alpha_3(Y_t = 3) + \epsilon_{t+1}, \quad Y_t > 0$$

where $\epsilon_{t+1} \sim \mathcal{N}(0, \sigma)$ is unknown to the individual until $t + 1$.

(a) Estimate the transition parameters α and σ . (Truth: $\alpha = [0.0, -0.75, 0.05, -0.25]$, $\sigma = 1$)

Coefficient	Value
α_0	0.0165
α_1	-0.7540
α_2	0.0345
α_3	-0.2517
σ	1.0067

(b) Use the estimated $\hat{\alpha}$ and $\hat{\sigma}$ to construct a $10 \times 10 \times 3$ discrete transition matrix, where

the first dimension is future X , the second is current X , and the third is work choice. Use a grid of X values at the inverse normal CDF evaluated at $[0.05, 0.15, \dots, 0.95]$, and create cutoff points for grid assignment.

(c) Create a discretized version of X_{1t} using the bins above. Use discretized states only for expected future values; contemporaneous flow utility is still a function of X_{1t} .

(d) Estimate the dynamic discrete choice model including the discount factor. Future expected utility depends on current choice and current state; use the transition matrix to integrate over the future X distribution. True parameters: coefficients on X 's (constant, gender, health) for three choices $[-0.75, 0.25, 0.25]$, $[-0.25, 0.5, 0.5]$, $[-0.85, 0.75, 0.75]$; switching cost 0.4 (enters negatively); discount factor 0.8.

	Coefficient	Estimate
Flex-Time		
	β_0	-0.789
	β_{Gender}	-0.501
	β_{Health}	0.083
Part-Time		
	β_0	-0.501
	β_{Gender}	0.607
	β_{Health}	0.199
Full-Time		
	β_0	-0.882
	β_{Gender}	0.746
	β_{Health}	0.325
	δ	0.204
	β_{DF}	0.822

My estimates are mostly terrible. I am sorry. I was not able to get them where I wanted while balancing all other work. I will work at it when I have more time since this is directly applicable to my work.

(e) What variation in the data allows the discount factor to be identified?

The expectation over future health states identifies the discount factor. That is, with some rational expectation of future health states, we can adequately discount the future.